

Ji Jungmin

Senior Backend Engineer

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SUMMARY

Senior backend engineer with 18 years in large-scale commerce, now pairing that depth with AI/full-stack indie products. End-to-end ownership from architecture and technical decisions through team execution and production operations.

- **16 years** at Auction/Gmarket; **3 years** leading the Item Engineering team (up to 19 engineers)
- Full-lifecycle architecture, build, and operations of a product platform at **300M+ records** · **thousands of TPS**
- Legacy modernization specialist — MSA migration, MySQL sharding, MSSQL→MongoDB migration, and zero-downtime large-scale data migration
- Operating **6 indie products across AI / full-stack / backend automation**: pet (PetCut) & human (SuppleCut) AI supplement analysis, a live commerce store (jullyssy.shop), stock-prediction automation (KronosStock), a Slack job-queue hub, and a B2B hotel-ordering PWA (Pizza of Legend)

SKILLS

CATEGORY	TECHNOLOGIES
Backend	Java 21 (primary), Spring Boot 3.x, C#.NET, Python 3.11
Frontend / Mobile	Next.js 14 (App Router), Flutter / Dart
DB / Storage	MySQL (Sharding), MSSQL, MongoDB (WiredTiger), PostgreSQL, Redis, Supabase
Messaging / Pipeline	Apache Kafka, Apache Spark (Streaming), APScheduler
Infra / Ops	Kubernetes (Private Cloud), Docker Compose, Vercel, Cloudflare Pages, Supabase Edge Functions (Deno/TS), GitHub Actions (CI/CD), Hetzner VPS, Grafana, Prometheus, DataDog
Middleware	Apache ShardingSphere-JDBC

CATEGORY	TECHNOLOGIES
AI / ML / LLM	Google Gemini API, Anthropic Claude API, cost-tiered multi-model routing, circuit-breaker & caching patterns, PyTorch CPU inference (Kronos-mini vendoring), AI-assisted development (Claude Code)
Payments	TossPayments, Google Play Billing (IAP)
Collaboration / Testing	Git, Jira, Postman, Playwright (E2E)
Architecture	MSA, CQRS, Event-Driven, Saga Pattern, DDD, BFF/Proxy, Idempotent Job Queue

EXPERIENCE

Gmarket Inc.

Item Engineering Team Lead · Sep 2011 – Nov 2024 (13 yrs 3 mos)

Led the core domain team owning the entire Gmarket/Auction product registration, edit, and query platform. Team of up to 19. Left via a voluntary retirement program prompted by changing business conditions.

Product Description MongoDB Migration & MSA Transition

Oct 2023 – Aug 2024 · Role: Architecture design · Technical decisions · Project lead

- **90% storage reduction** (13TB → 1.33TB): MSSQL dual-write → MongoDB BSON + Snappy/Zlib compression + MD5 deduplication
- Migrated **1B+ records / ~13TB** safely via a **zero-downtime 5-phase transition**
- Designed for Write **50–150 TPS / ~20ms avg latency**, Read **5,500 TPS steady / 16,000 TPS peak**
- Selected MongoDB after a quantitative comparison of 5 databases (PostgreSQL, Elasticsearch, Cassandra, Redis, MongoDB) — led the team's technical decision
- Split into 3 collections (`descriptionMeta` / `description` / `descriptionUpdateHistory`) to minimize update overhead
- Established an RDB-fallback strategy during MongoDB outages to preserve service continuity

ESM+ Product Registration/Edit Backend Re-architecture

2022 – Jun 2024 · Role: Project lead · Backend architect

- **~30% productivity gain (est.)**: consolidated dual legacy/new systems into a single unified system; one shared module deployed across both Gmarket and Auction (estimated from eliminated per-site duplicate development effort)
- **~25% QA-time reduction (est.)**: module-level independent testing narrowed QA scope

- Completed **zero-downtime migration of 100M of 300M total records (30%)** — Facade pattern + progressive 3-stage transition
- Decomposed a monolith into 3 domain modules (core product info / shipping / options); Saga pattern + compensating transactions for data consistency
- Built real-time consistency-validation batches + auto-recovery; Kafka-based event sync
- Ran a 2.5-year roadmap, stakeholder alignment, and dev standards (code review, API design guides)

Sharded MySQL Cluster Product Query Platform

2022 · **Role: System architect · Tech lead**

- **Stable 8,000 TPS under event traffic:** resolved single-MSSQL bottleneck (CPU/Memory limits during events) with a 4-shard MySQL cluster
- **Eliminated SPOF:** independent per-shard operation blocks single points of failure and prevents service-wide cascading outages
- Evenly distributed **~300M product records** across 4 shards (Modular Sharding strategy)
- Chose MySQL + ShardingSphere-JDBC after a PostgreSQL vs MySQL analysis — license-cost savings vs MSSQL
- Designed a custom near-real-time sync pipeline (`MSSQL → Kafka → MySQL` , Spark Job + Spring Boot Consumer) where standard CDC was not viable
- **Improved JVM cold start from 3s → 20–40ms** (root-caused via DataDog instrumentation, applied Spring WarmUp)

Gmarket/Auction Product 2.0 (Product DB Integration & Registration Overhaul)

2017 · **Role: Backend lead · DB modeling · Project management**

- Unified Gmarket/Auction's dual product DBs and registration/edit systems — improved seller experience and internal operational efficiency
- Led full DB modeling and service architecture; designed an extensible structure for multi-channel and varied sales types (single/multi/bundle)
- **Led 8 external developers**, reviewed deliverable quality, managed schedule

ESM+ Unified Seller Tools TF

2011 – 2012 · **Role: Backend developer**

- Joined the Auction/Gmarket service integration (ESM+) task force; designed and integrated Auction product-domain APIs
- Secured cross-service data compatibility and removed redundancy

eBay Auction Co.

Item Team · May 2008 – Sep 2011 (3 yrs 5 mos)

- Backend/frontend development and maintenance for Auction product registration/edit/query
- Built the product auction system and a bulk-integration OpenAPI for sellers

TmaxSoft Co.

Staff Engineer · Oct 2006 – Mar 2008 (*1 yr 6 mos*)

- Citibank PNC project: GIRO/CMS auto-transfer registration/change/query development, monthly fee batch jobs
 - Voluntarily moved on to gain large-scale commerce domain experience (joined Auction)
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| SIDE PROJECTS (INDIE PRODUCT DEVELOPMENT)

Using 18 years of backend experience plus AI-assisted development (Claude Code) to personally design, build, ship, and operate full-stack / AI / automation products. Applies senior backend depth across many domains — payment PG integration, multi-LLM routing, backend proxy & circuit-breaker patterns, and trading automation.

PetCut — AI Pet Food + Supplement Combo Analyzer (US pet market)

2025 – ongoing · **Solo Developer (Architect · Backend · Frontend)** · github.com/kaos1025/PetCut

- **Supabase Edge Function (Deno/TS) proxy backend:** fully removes LLM API keys from the client + IP/device rate limiting + an "S8" circuit breaker (spend cap) + catalog/cache pre-lookup to minimize LLM calls
- **Domain differentiation:** not single-product scanning but "food + supplement combo" analysis — simultaneous evaluation of weight-based toxicity thresholds (D3, Iron, Ca), ingredient overlap, and mechanism conflicts (zero direct competitors)
- **Strategic BM pivot:** the L1 era (Claude \$1.99 IAP paid report; live-validated on the Android Internal Track at versionCode 7 with 321 tests) was intentionally deprecated → redefined V1 as web-only with a free Chewy/Amazon affiliate model (decided from objective market variables D1–D6)
- **Stack:** Flutter (V1.1 app), Supabase Edge Functions (Deno/TS), Supabase Postgres, Gemini Flash, Cloudflare Pages, Playwright E2E

SuppleCut — AI Supplement Stack Analyzer (sister product)

2025 · **Solo Developer (full-stack)** · supplecut.com · Google Play closed beta

- **Multi-LLM routing:** Gemini Flash (free, deterministic analysis — OCR, dedup, UL calculation) → Claude Sonnet (paid IAP \$1.99, narrative explanation) — "Decisions by Engine, Explanations by AI" to separate cost from quality
- **Passed 20/20 pharmacist-reviewed test cases** (3–12 product cases, including edge cases like 7-product GABA pathway overlap)
- **Concept → Google Play closed beta in one month**, Google Play Billing IAP integrated (verified on real devices)

- **Stack:** Flutter / Dart, Google Gemini API, Anthropic Claude API, Google Play Billing, Cloudflare Pages

Jullyssy Mall — Women's Fashion D2C Store (client project)

2025 – ongoing · **Solo Developer (full-stack)** · jullyssy.shop (live; PG review in progress)

- **Concept → live in ~1 month** — a real client project migrating a Naver Smart Store into an independent D2C store; bulk-imported existing products via the Naver Commerce API
- **Payment/order reliability:** TossPayments integration (card, bank transfer, KakaoPay, NaverPay) + raw-response JSONB retention (for dispute handling); atomic stock decrement via RPC + Order Snapshot pattern for concurrency and order accuracy
- **Automated SEO backend pipeline:** Supabase Edge Function (seo-generate-worker) + pg_cron schedule + dead-letter queue + stuck-recover logic for automated metadata generation and an approval workflow
- **Full commerce operations:** Kakao OAuth login, coupons (fixed/percentage, minimum-order), reviews (rating + image + height/weight/size tag-summary RPC), returns/exchanges, event categories, admin (products, orders, tracking numbers, banners) — scaled progressively across 53 migrations
- **Operational hardening:** Sentry error monitoring, 8 passing Playwright E2E specs (auth, order, cancel, coupon, guest cart, admin), RLS + service-role separation
- **Stack:** Next.js 14 (App Router), Tailwind + shadcn/ui, Supabase (PostgreSQL + Auth + Storage + Edge Functions), TossPayments, Sentry, Vercel, Playwright E2E

KronosStock — Korean Stock Prediction + Dry-Run Automation (AAAI 2026 Kronos-mini)

Solo Developer (Backend + ML Integration) · github.com/kaos1025/KronosStock

- **Productionizing a research model:** vendors AAAI 2026 Kronos-mini directly (`inference/vendor_kronos.sh`) into a KOSPI/KOSDAQ daily price-prediction pipeline. Persistent HuggingFace Hub weight cache, KronosForecaster loaded once per process (singleton, for CPU performance), fixed model↔tokenizer mapping. A personal, non-commercial learning/experimentation tool.
- **Backend + safe automation design:** FastAPI dashboard + Redis OHLCV buffering + APScheduler 4-cron jobs (08:50 / 09:30 / 12:00 / 15:20 KST, with `exchange-calendars` XKRX holidays auto-applied). Runs the full Forecast → Signal → Paper Order → Digest flow in dry-run mode by default — blocks real broker order-API calls, updating only a paper portfolio snapshot.
- **External-API ops + testing/CI-CD:** 3-tier data fallback (Korea Investment & Securities Open API via python-kis + FinanceDataReader + pykrx), rate-limit aware (live 20 req/s · mock 5 req/s) + 24h token disk cache. Unit-testable without network via model stubs + fakeredis; auto-deploy to a Hetzner CX22 VPS via GitHub Actions (`deploy-vps.yml`).
- **Stack:** Python 3.11, FastAPI, Redis, APScheduler, PyTorch CPU, python-telegram-bot, python-kis, exchange-calendars (XKRX), Docker Compose, GitHub Actions, Hetzner VPS

Other Projects

-  **Slack Webhook Hub** (github.com/kaos1025/slack-webhook-hub) — a Slack command hub with channel-based routing and a swappable execution backend (routine/noop/worker). A retry-safe idempotent `command_jobs` queue (Supabase/PostgREST) plus a worker that runs agent jobs with worktree isolation and artifact logging. Next.js / TypeScript.

- 🍕 **Pizza of Legend Express** — a **real B2B-delivered** QR-ordering PWA for foreign guests at 5 hotels in Yeongjong, Korea. No sign-up, no app install, order via QR scan; multilingual (EN / 中文 / 日本語); Supabase Realtime order operations + Telegram/push notifications (configurable directly from the owner's admin UI). Next.js 14 / next-intl / Supabase.

PROJECTS SUMMARY

PROJECT	HIGHLIGHTS	STACK
MongoDB Migration & MSA	90% storage cut (13TB→1.33TB), 1B+ records migrated zero-downtime	Java 21, Spring Boot 3.2, MongoDB
Sharded MySQL Query Platform	Stable 8,000 TPS, SPOF removed, JVM 3s→40ms	MySQL Cluster, ShardingSphere, Kafka, Spark
ESM+ Re-architecture	~30% productivity ↑ (est.), ~25% QA-time ↓ (est.), 100M records migrated zero-downtime	Java, Spring Boot, Kafka, MSSQL
Product 2.0	Gmarket/Auction DB unification, led 8 external devs	C#.NET, MSSQL
PetCut (Indie)	Edge Function proxy + S8 circuit breaker + catalog cache, food+supplement combo analysis	Flutter, Supabase Edge Function, Gemini
SuppleCut (Indie)	Multi-LLM routing, Google Play beta, IAP \$1.99 validated, 20/20 pharmacist review	Flutter, Gemini, Claude, Play Billing
Jullyssy Mall (Indie)	~1-month solo full-stack, TossPayments + automated SEO Edge Function, 53 migrations, live	Next.js 14, Supabase, TossPayments, Sentry
KronosStock (Indie)	AAAI 2026 ML model vendoring + dry-run automation, Redis/APScheduler/VPS CI-CD	Python, FastAPI, PyTorch, Redis, GitHub Actions
Slack Webhook Hub (Indie)	Channel routing + idempotent job queue + swappable execution backend (worker)	Next.js, TypeScript, Supabase/PostgREST
Pizza of Legend (B2B)	QR-ordering PWA delivered to 5 hotels, multilingual + Realtime ops + notifications	Next.js 14, next-intl, Supabase Realtime

EDUCATION & CERTIFICATIONS

Inha University — B.S. in Computer Science · 1999 – 2006

- Engineer Information Processing (정보처리기사)

- 🏆 **Gmarket DNA Awards, H1 2023** — recognized for domain expertise and proactive problem-solving